IEDA 4000H

Tutorial 7

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1. Key Knowledge Points Review
2. Example Problems

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2. Example Problems

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- Lagrangian Function and Lagrangian Duality
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Robust Optimization (Lecture 11)

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- Robust Counterpart

1. Key Knowledge Points Review
2. Example Problems

Problem 1

Judge and verify that the following functions are convex/concave/both convex and concave/neither convex nor concave on their domains.

(1) $f(x) = e^x - \frac{1}{6}(1+x)^3,\ x \in \mathbb{R}$;

(2) $f(x_1, x_2) = \frac{x_1^2}{x_2} - 1 \leq 0,\ x_1 \in \mathbb{R},\ x_2 > 0$;

(3) $f(\mathbf{x}) = \min\limits_{1 \leq i \leq m} \mathbf{a}_i^\top \mathbf{x},\ \mathbf{x} \in \mathbb{R}^n$, where $\{\mathbf{a}_i \in \mathbb{R}^n\}_{1 \leq i \leq m}$ are given.

Solution 1

(1): Convex. Since $f''(x) = e^x - (x+1) \geq 0$ for all $x \in \mathbb{R}$ by $e^x \geq 1 + x$ (equality only at $x = 0$).

(2): Convex. Since

$$\nabla^2 f(x_1, x_2) = \begin{pmatrix} \frac{2}{x_2} & -\frac{2x_1}{x_2^2} \\ -\frac{2x_1}{x_2^2} & \frac{2x_1^2}{x_2^3} \end{pmatrix} = \frac{2}{x_2^3} \begin{pmatrix} x_2^2 & -x_1 x_2 \\ -x_1 x_2 & x_1^2 \end{pmatrix} \succeq 0, \quad \forall x_2 > 0.$$

(3): Concave. It is the pointwise minimum of affine functions (both convex and concave), hence concave.

Problem 2

Consider the optimization problem

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^n} \quad & \frac{1}{2}\|\mathbf{x}\|^2 \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b}, \\ & \|\mathbf{x}-\mathbf{c}\|_2^2 \le 1, \end{aligned}$$

where $A \in \mathbb{R}^{m\times n}$, $\mathbf{b} \in \mathbb{R}^m$, and $\mathbf{c} \in \mathbb{R}^n$ are given. Assume that the feasible set is non-empty.

(1) Find the Lagrangian dual.

(2) Write down the KKT conditions.

Solution 2 (1)

(1): The Lagrangian $L(\mathbf{x}, \lambda, \mu)$ is:

$$\begin{aligned} L(\mathbf{x}, \lambda, \mu) &= \frac{1}{2}\|\mathbf{x}\|^2 + \lambda^T(\mathrm{A}\mathbf{x} - \mathbf{b}) + \mu(\|\mathbf{x} - \mathbf{c}\|_2^2 - 1) \\ &= \frac{1}{2}\|\mathbf{x}\|^2 + \lambda^T\mathrm{A}\mathbf{x} - \lambda^T\mathbf{b} + \mu(\|\mathbf{x}\|^2 - 2\mathbf{c}^T\mathbf{x} + \|\mathbf{c}\|^2 - 1) \\ &= \left(\frac{1}{2} + \mu\right)\|\mathbf{x}\|^2 + (\lambda^T\mathrm{A} - 2\mu\mathbf{c}^T)\mathbf{x} - \lambda^T\mathbf{b} + \mu(\|\mathbf{c}\|^2 - 1), \quad \lambda \in \mathbb{R}^m,\ \mu \geq 0. \end{aligned}$$

To find the dual function, we minimize L with respect to $\mathbf{x}$, consider the dual function

$$\phi(\lambda, \mu) = \inf_{\mathbf{x} \in \mathbb{R}^n} L(\mathbf{x}, \lambda, \mu).$$

Solution 2 (1)

The gradient of L with respect to $\mathbf{x}$ is:

$$\begin{aligned}\nabla_{\mathbf{x}} L &= 2\left(\frac{1}{2}+\mu\right)\mathbf{x} + \mathrm{A}^T\lambda - 2\mu\mathbf{c} \\ &= (1+2\mu)\mathbf{x} + \mathrm{A}^T\lambda - 2\mu\mathbf{c}\end{aligned}$$

Setting the gradient to zero to find the minimizer $\mathbf{x}^*$:

$$\begin{aligned}(1+2\mu)\mathbf{x}^* &= 2\mu\mathbf{c} - \mathrm{A}^T\lambda \\ \mathbf{x}^* &= \frac{2\mu}{1+2\mu}\mathbf{c} - \frac{1}{1+2\mu}\mathrm{A}^T\lambda\end{aligned}$$

For the infimum to be finite, we need the coefficient of the quadratic term to be positive, i.e., $1+2\mu > 0$. This implies $\mu > -\frac{1}{2}$. Since $\mu \geq 0$ from the dual feasibility requirement, this condition is automatically satisfied.

Solution 2 (1)

Substituting the optimal $\mathbf{x}^*$ back into the Lagrangian gives the dual function $\phi(\lambda, \mu)$:

$$\begin{aligned}\phi(\lambda, \mu) &= L(\mathbf{x}^*, \lambda, \mu) \\ &= -\frac{1}{2(1+2\mu)}\|\mathrm{A}^T\lambda - 2\mu\mathbf{c}\|^2 - \lambda^T\mathbf{b} + \mu(\|\mathbf{c}\|^2 - 1)\end{aligned}$$

Therefore, the dual problem is to maximize the dual function subject to the dual feasibility constraint $\mu \geq 0$:

$$\begin{aligned}\max_{\lambda\in\mathbb{R}^m,\ \mu\in\mathbb{R}} \quad & -\tfrac{1}{2(1+2\mu)}\|\mathrm{A}^T\lambda - 2\mu\mathbf{c}\|^2 - \lambda^T\mathbf{b} + \mu(\|\mathbf{c}\|^2 - 1) \\ \text{s.t.} \quad & \mu \geq 0.\end{aligned}$$

Solution 2 (2)

(2): The KKT conditions are

$$
\begin{aligned}
\text{Primal feasibility:}\quad & A\mathbf{x} = \mathbf{b}, \quad \|\mathbf{x}-\mathbf{c}\|_2^2 \le 1, \\
\text{Dual feasibility:}\quad & \mu \ge 0, \\
\text{Stationarity:}\quad & (1+2\mu)\mathbf{x} + A^T\lambda - 2\mu\mathbf{c} = 0, \\
\text{Complementary slackness:}\quad & \mu\left(\|\mathbf{x}-\mathbf{c}\|_2^2 - 1\right) = 0.
\end{aligned}
$$

Problem 3

Compute the conjugate of the following functions:

(1) $f(x) = e^x,\ x \in \mathbb{R}$;

(2) $f(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0; \end{cases}$

(3) $f(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2,\ \mathbf{x} \in \mathbb{R}^n$.

Solution 3 (1)

(1): We need to compute

$$f^*(y) = \sup_{x\in\mathbb{R}} \left(yx - e^x\right).$$

To find the supremum, consider the function $g(x) = yx - e^x$. This is differentiable, so we find critical points by setting the derivative to zero:

$$g'(x) = y - e^x = 0 \implies e^x = y \implies x = \ln y,$$

provided $y > 0$ (since $x \in \mathbb{R}$ requires $y > 0$ for the logarithm to be defined).

Solution 3 (1)

For $y \leq 0$, we analyze separately.

- Case $y = 0$: Now $g(x) = -e^x$. The supremum is $\sup_{x\in\mathbb{R}}(-e^x) = 0$ (achieved as $x \to -\infty$).

  Note that $\lim_{y\to 0^+}(y\ln y - y) = 0$ (using the convention $0\ln 0 = 0$), so the expression $y\ln y - y$ extends continuously to $y = 0$.
- Case $y < 0$: Here, $g'(x) = y - e^x < 0$ for all x (since $y < 0$ and $-e^x < 0$). Thus, $g(x)$ is strictly decreasing. As $x \to -\infty$, $yx \to +\infty$ (since $y < 0$) and $-e^x \to 0$, so $g(x) \to +\infty$. Therefore, $f^*(y) = +\infty$.

Therefore, $f^*(y) = +\infty$. Combining cases:

$$f^*(y) = \begin{cases} y\ln y - y & y \geq 0, \\ +\infty & y < 0, \end{cases}$$

with the convention that $0\ln 0 = 0$.

Solution 3 (2)

(2): Note that f is the sign function (with $\operatorname{sgn}(0) = 0$), which is not convex but extended-real-valued in the conjugate context. We compute

$$f^*(y) = \sup_{x \in \mathbb{R}} \left(yx - f(x) \right).$$

Since f is piecewise constant, we evaluate $g(x) = yx - f(x)$ over the pieces $x < 0$, $x = 0$, and $x > 0$.

Solution 3 (2)

- For $x > 0$: $g(x) = yx - 1$. If $y > 0$, this goes to $+\infty$ as $x \to +\infty$; if $y < 0$, it goes to $-\infty$ as $x \to +\infty$, and the max on $(0, +\infty)$ is approached as $x \to 0^+$, giving $g(x) \to -1$; if $y = 0$, $g(x) = -1$.
- For $x = 0$: $g(0) = 0 - 0 = 0$.
- For $x < 0$: $g(x) = yx - (-1) = yx + 1$. If $y < 0$, this goes to $+\infty$ as $x \to -\infty$ (since $yx > 0$ and large); if $y > 0$, it goes to $-\infty$ as $x \to -\infty$, and the max on $(-\infty, 0)$ is approached as $x \to 0^-$, giving $g(x) \to +1$; if $y = 0$, $g(x) = +1$.

Solution 3 (2)

Now, combine to find the global supremum:

- Case $y > 0$: On $x > 0$, $g(x) \to +\infty$; on $x < 0$, max is $+1$; at $x = 0$, 0. Thus, $\sup g(x) = +\infty$.
- Case $y < 0$: On $x < 0$, $g(x) \to +\infty$; on $x > 0$, max is -1; at $x = 0$, 0. Thus, $\sup g(x) = +\infty$.
- Case $y = 0$: On $x > 0$, $g(x) = -1$; at $x = 0$, 0; on $x < 0$, $g(x) = +1$. Thus, $\sup g(x) = 1$ (achieved for all $x < 0$).

Therefore:

$$f^*(y) = \begin{cases} 1 & y = 0, \\ +\infty & y \neq 0. \end{cases}$$

This is the conjugate of the (non-convex) sign function, resulting in an indicator-like function (finite only at $y = 0$).

Solution 3 (3)

(3): We compute

$$f^*(\mathbf{y}) = \sup_{\mathbf{x}\in\mathbb{R}^n} \left(\mathbf{y}^\top \mathbf{x} - \frac{1}{2}\|\mathbf{x}\|^2 \right).$$

Consider $g(\mathbf{x}) = \mathbf{y}^\top \mathbf{x} - \frac{1}{2}\|\mathbf{x}\|^2$, which is differentiable. The gradient is

$$\nabla g(\mathbf{x}) = \mathbf{y} - \mathbf{x} = \mathbf{0} \implies \mathbf{x} = \mathbf{y}.$$

The Hessian is $-\mathrm{I}$ (negative definite), so $\mathbf{x} = \mathbf{y}$ is a maximum. Substitute:

$$g(\mathbf{y}) = \mathbf{y}^\top \mathbf{y} - \frac{1}{2}\|\mathbf{y}\|^2 = \|\mathbf{y}\|^2 - \frac{1}{2}\|\mathbf{y}\|^2 = \frac{1}{2}\|\mathbf{y}\|^2.$$

As $\|\mathbf{x}\| \to \infty$ in directions away from $\mathbf{y}$, $g(\mathbf{x}) \to -\infty$ (quadratic term dominates), confirming the supremum is finite and achieved at $\mathbf{x} = \mathbf{y}$. Thus:

$$f^*(\mathbf{y}) = \frac{1}{2}\|\mathbf{y}\|^2, \quad \mathbf{y} \in \mathbb{R}^n.$$

This shows that the conjugate of the quadratic is itself (self-dual).

Problem 4

Consider the linear robust constraint

$$(\bar{\mathbf{a}} + \mathrm{P}\mathbf{z})^\top \mathbf{x} \leq b, \quad \forall \mathbf{z} \in \mathcal{Z},$$

where $\bar{\mathbf{a}} \in \mathbb{R}^n$ is the known nominal vector, $b \in \mathbb{R}$ and $\mathrm{P} \in \mathbb{R}^{n \times m}$ are given, $\mathcal{Z} \subseteq \mathbb{R}^m$ is the uncertainty set.
Define

$$\mathcal{Z}_1 = \left\{\mathbf{z} : \|\mathbf{z}\|_2 \leq 1,\ \mathbf{z} \in \mathbb{R}^m\right\}$$

and

$$\mathcal{Z}_2 = \left\{\mathbf{z} : \|\mathbf{z} - \mathbf{c}\|_2 \leq 1,\ \mathbf{z} \in \mathbb{R}^m\right\},$$

where $\mathbf{c} \in \mathbb{R}^m$ is a given vector with $\|\mathbf{c}\|_2 < 2$. Let $\mathcal{Z} = \mathcal{Z}_1 \cap \mathcal{Z}_2$. Find the robust counterpart.

Solution 4

Let $\mathbf{d} = P^\top \mathbf{x}$. The key is to compute $\sup_{\mathbf{z}\in\mathcal{Z}_1\cap\mathcal{Z}_2} \mathbf{d}^\top \mathbf{z}$ using convex conjugates. Define the indicator function of a set $S \subseteq \mathbb{R}^m$ as

$$I_S(\mathbf{z}) = \begin{cases} 0 & \mathbf{z} \in S, \\ +\infty & \mathbf{z} \notin S. \end{cases}$$

The convex conjugate of I_S is the support function of S:

$$I_S^*(\mathbf{y}) = \sup_{\mathbf{z}\in S} \mathbf{y}^\top \mathbf{z} =: \sigma_S(\mathbf{y}).$$

Since $\mathcal{Z} = \mathcal{Z}_1 \cap \mathcal{Z}_2$, the indicator satisfies $I_{\mathcal{Z}}(\mathbf{z}) = I_{\mathcal{Z}_1}(\mathbf{z}) + I_{\mathcal{Z}_2}(\mathbf{z})$. The conjugate of a sum of functions $f + g$ is the infimal convolution of their conjugates:

$$(f+g)^*(\mathbf{y}) = \inf_{\mathbf{u}+\mathbf{v}=\mathbf{y}} f^*(\mathbf{u}) + g^*(\mathbf{v}).$$

Solution 4

Thus,

$$\sigma_{\mathcal{Z}}(\mathbf{d}) = I^*_{\mathcal{Z}}(\mathbf{d}) = (I_{\mathcal{Z}_1} + I_{\mathcal{Z}_2})^*(\mathbf{d}) = \inf_{\mathbf{u}+\mathbf{v}=\mathbf{d}} I^*_{\mathcal{Z}_1}(\mathbf{u}) + I^*_{\mathcal{Z}_2}(\mathbf{v}) = \inf_{\mathbf{u}+\mathbf{v}=\mathbf{d}} \sigma_{\mathcal{Z}_1}(\mathbf{u}) + \sigma_{\mathcal{Z}_2}(\mathbf{v}).$$

Now, compute the support functions explicitly:
For $\mathcal{Z}_1 = \{\mathbf{z} : \|\mathbf{z}\|_2 \le 1\}$ (unit ball centered at $\mathbf{0}$):

$$\sigma_{\mathcal{Z}_1}(\mathbf{u}) = \sup_{\|\mathbf{z}\|_2 \le 1} \mathbf{u}^\top \mathbf{z} = \|\mathbf{u}\|_2,$$

by Cauchy-Schwarz (achieved at $\mathbf{z} = \mathbf{u}/\|\mathbf{u}\|_2$ if $\mathbf{u} \ne \mathbf{0}$).
For $\mathcal{Z}_2 = \{\mathbf{z} : \|\mathbf{z} - \mathbf{c}\|_2 \le 1\}$ (unit ball centered at $\mathbf{c}$):

$$\sigma_{\mathcal{Z}_2}(\mathbf{v}) = \sup_{\|\mathbf{w}\|_2 \le 1} \mathbf{v}^\top (\mathbf{c} + \mathbf{w}) = \mathbf{v}^\top \mathbf{c} + \sup_{\|\mathbf{w}\|_2 \le 1} \mathbf{v}^\top \mathbf{w} = \mathbf{v}^\top \mathbf{c} + \|\mathbf{v}\|_2.$$

Solution

Therefore,

$$\sigma_{\mathcal{Z}}(\mathbf{d}) = \inf_{\mathbf{u}+\mathbf{v}=\mathbf{d}} \left(\|\mathbf{u}\|_2 + \mathbf{v}^\top \mathbf{c} + \|\mathbf{v}\|_2 \right).$$

The robust constraint becomes

$$\bar{\mathbf{a}}^\top \mathbf{x} + \inf_{\mathbf{u}+\mathbf{v}=\mathrm{P}^\top \mathbf{x}} \left(\|\mathbf{u}\|_2 + \|\mathbf{v}\|_2 + \mathbf{c}^\top \mathbf{v} \right) \leq b.$$

This infimum is over affine constraints, so by introducing auxiliary variables $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$, the constraint is equivalent to the existence of $\mathbf{u}, \mathbf{v}$ such that

$$\mathbf{u} + \mathbf{v} = \mathrm{P}^\top \mathbf{x}, \quad \bar{\mathbf{a}}^\top \mathbf{x} + \|\mathbf{u}\|_2 + \|\mathbf{v}\|_2 + \mathbf{c}^\top \mathbf{v} \leq b.$$

This is a convex constraint, as it involves a sum of Euclidean norms (convex) and affine terms.

Solution 4

It is second-order cone representable (SOCP): introduce scalars $t_1 \geq \|\mathbf{u}\|_2$ and $t_2 \geq \|\mathbf{v}\|_2$ (each enforceable by a second-order cone constraint $\|\mathbf{u}\|_2 \leq t_1$, etc.), yielding

$$\bar{\mathbf{a}}^\top \mathbf{x} + t_1 + t_2 + \mathbf{c}^\top \mathbf{v} \leq b, \quad \mathbf{u} + \mathbf{v} = \mathrm{P}^\top \mathbf{x}, \quad t_1 \geq \|\mathbf{u}\|_2, \quad t_2 \geq \|\mathbf{v}\|_2, \quad t_1, t_2 \geq 0.$$

Thank you for listening !

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